Module 19

Entity-Relationship Model/2

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Module Recap

- Design Process for Database Systems
- ER Model for real world representation with entities, entity sets, attributes, and relationships

Module Objectives

- To illustrate ER Diagram notation for ER Models
- To explore translation of ER Models to Relational Schemas

Module Outline

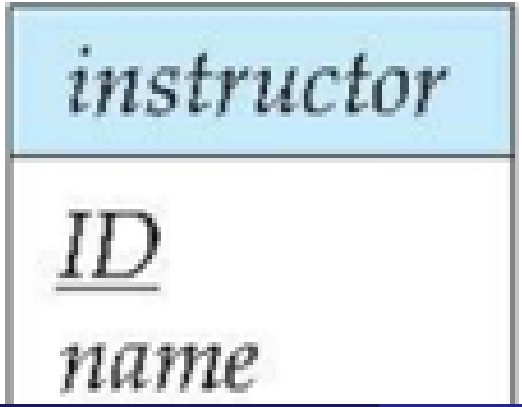
- ER Diagram
- ER Model to Relational Schema

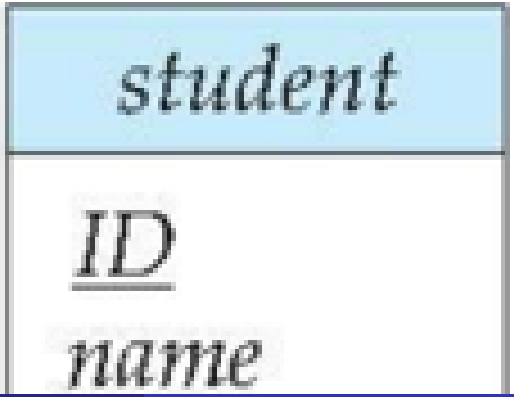
ER Diagram

Entity Sets

Entities can be represented graphically as follows:

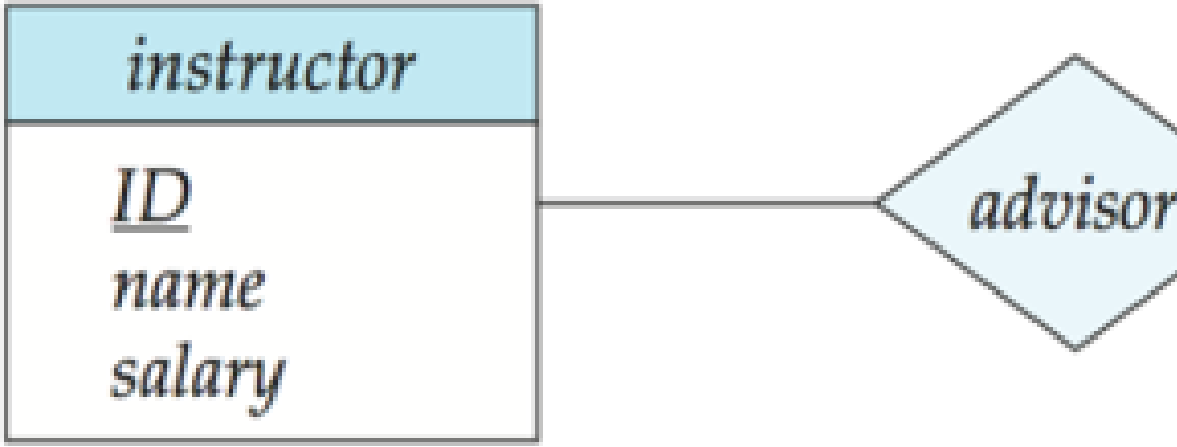
- Rectangles represent entity sets.
- Attributes are listed inside entity rectangle.
- Underline indicates primary key attributes.



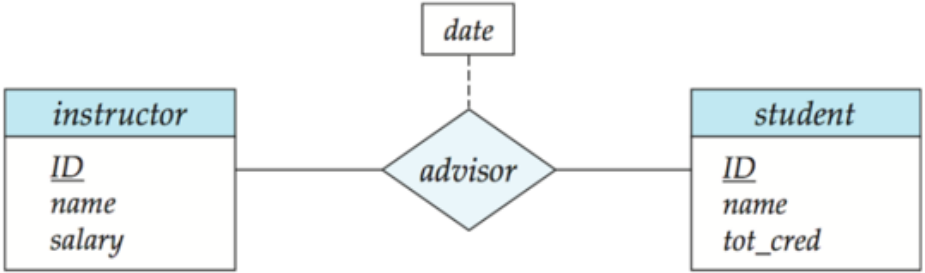


Relationship Sets

- Diamonds represent relationship sets.

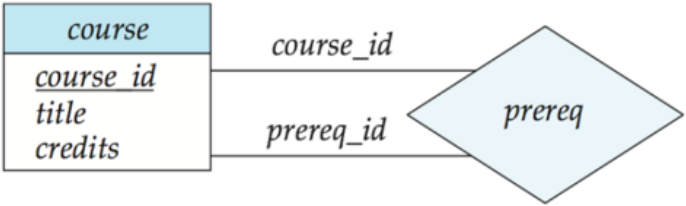


Relationship Sets with Attributes



Roles

- Entity sets of a relationship need not be distinct Each occurrence of an entity set plays a 'role' in the relationship
- The labels 'course id' and 'prereq id' are called **roles**.



Cardinality Constraints

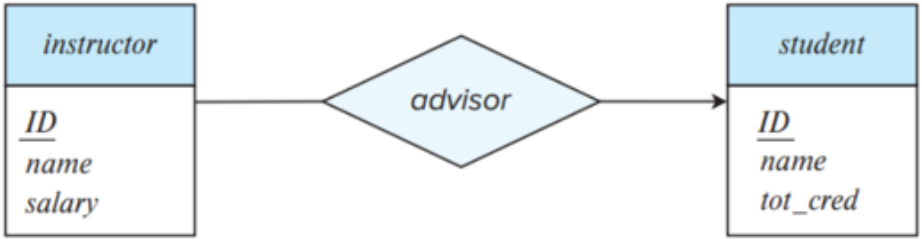
- We express cardinality constraints by drawing either a directed line ($\rightarrow$), signifying 'one,' or an undirected line ($-$), signifying 'many,' between the relationship set and the entity set.
- **One-to-one relationship between an instructor and a student**:
  - A student is associated with at most one instructor via the relationship advisor
  - An instructor is associated with at most one student via the relationship advisor

One-to-Many Relationship

- **one-to-many relationship between an instructor and a student**
- an instructor is associated with several (including 0) students via advisor
- a student is associated with at most one instructor via advisor

Many-to-One Relationships

- **many-to-one relationship between a student and an instructor**
- an instructor is associated with at most one student via advisor
- a student is associated with several (including 0) instructors via advisor

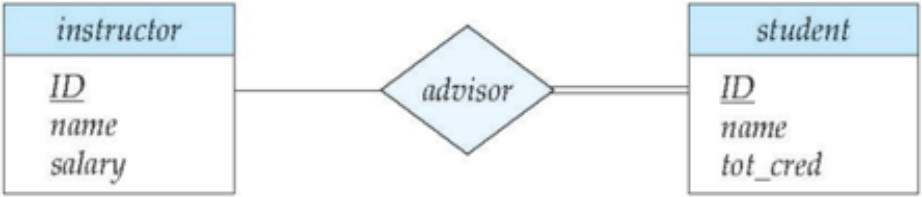


Many-to-Many Relationship

- An instructor is associated with several (possibly 0) students via advisor
- A student is associated with several (possibly 0) instructors via advisor

Total and Partial Participation

- **Total participation** (indicated by double line): every entity in the entity set participates in at least one relationship in the relationship set
  - participation of student in advisor relation is total
  - every student must have an associated instructor
- **Partial participation**: some entities may not participate in any relationship in the relationship set
  - Example: participation of instructor in advisor is partial



Notation for Expressing More Complex Constraints

- A line may have an associated minimum and maximum cardinality, shown in the form *l*..*h*, where *l* is the minimum and *h* the maximum cardinality
- A minimum value of 1 indicates total participation.
- A maximum value of 1 indicates that the entity participates in at most one relationship
- A maximum value of * indicates no limit.

Notation to Express Entity with Complex Attributes

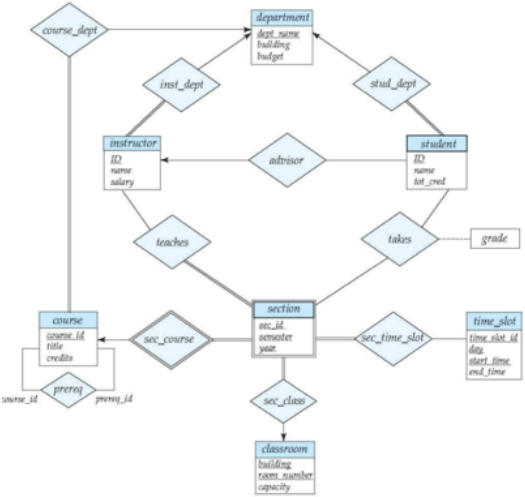
instructor

ID
name
    first_name
    middle_initial
    last_name
address
    street

Expressing Weak Entity Sets

- In ER diagrams, a weak entity set is depicted via a double rectangle
- We underline the discriminator of a weak entity set with a dashed line
- The relationship set connecting the weak entity set to the identifying strong entity set is depicted by a double diamond
- Primary key for section - (course_id, sec_id, semester, year)

ER Diagram for a University Enterprise



Partha Pratim Das

ER Model to Relational Schema

Reduction to Relation Schema

- Entity sets and relationship sets can be expressed uniformly as relation schemas that represent the contents of the database
- A database which conforms to an ER diagram can be represented by a collection of schemas
- For each entity set and relationship set there is a unique schema that is assigned the name of the corresponding entity set or relationship set
- Each schema has a number of columns (generally corresponding to attributes), which have unique names

Representing Entity Sets

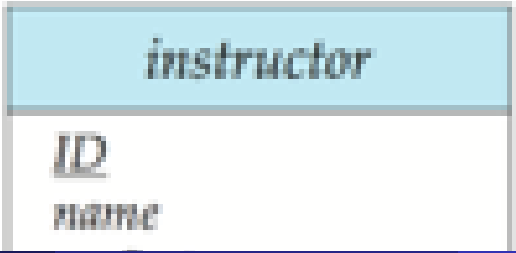
- A strong entity set reduces to a schema with the same attributes
  - `student(ID, name, tot_cred)`
- A weak entity set becomes a table that includes a column for the primary key of the identifying strong entity set
  - `section(course_id, sec_id, sem, year)`

Representing Relationship Sets

- A many-to-many relationship set is represented as a schema with attributes for the primary keys of the two participating entity sets, and any descriptive attributes of the relationship set.
- Example: schema for relationship set advisor
  - `advisor = (s_id, i_id)`

Representation of Entity Sets with Composite Attributes

- Composite attributes are flattened out by creating a separate attribute for each component attribute
- Example: given entity set `instructor` with composite attribute `name` with component attributes `first_name` and `last_name` the schema corresponding to the entity set has two attributes `name_first_name` and `name_last_name`
  - Prefix omitted if there is no ambiguity (`name_first_name` could be `first_name`)
- Ignoring multivalued attributes, extended `instructor` schema is
  - `instructor(ID, first_name, middle_initial, last_name, street_number, street_name, apt_number, city, state, zip_code, date_of_birth)`

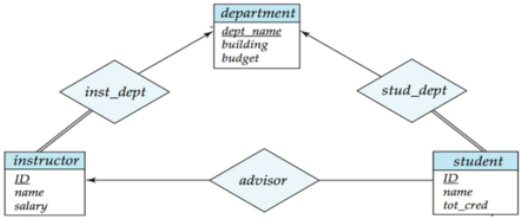


Representation of Entity Sets with Multivalued Attributes

- A multivalued attribute M of an entity E is represented by a separate schema E_M
- Schema E_M has attributes corresponding to the primary key of E and an attribute corresponding to multivalued attribute M
- Example: Multivalued attribute `phone_number` of `instructor` is represented by a schema: `inst_phone = (ID, phone_number)`
- Each value of the multivalued attribute maps to a separate tuple of the relation on schema E_M
- For example, an `instructor` entity with primary key 22222 and phone numbers 456-7890 and 123-4567 maps to two tuples: (22222, 456-7890) and (22222, 123-4567)

Redundancy of Schema

- Many-to-one and one-to-many relationship sets that are total on the many-side can be represented by adding an extra attribute to the 'many' side, containing the primary key of the 'one' side
- Example: Instead of creating a schema for relationship set `inst_dept`, add an attribute `dept_name` to the schema arising from entity set `instructor`



Redundancy of Schema (2)

- For one-to-one relationship sets, either side can be chosen to act as the 'many' side
- That is, an extra attribute can be added to either of the tables corresponding to the two entity sets
- If participation is partial on the 'many' side, replacing a schema by an extra attribute in the schema corresponding to the 'many' side could result in null values

Redundancy of Schema (3)

- The schema corresponding to a relationship set linking a weak entity set to its identifying strong entity set is redundant.
- Example: The `section` schema already contains the attributes that would appear in the `sec_course` schema

Module Summary

- Illustrated ER Diagram notation for ER Models
- Discussed translation of ER Models to Relational Schema

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